

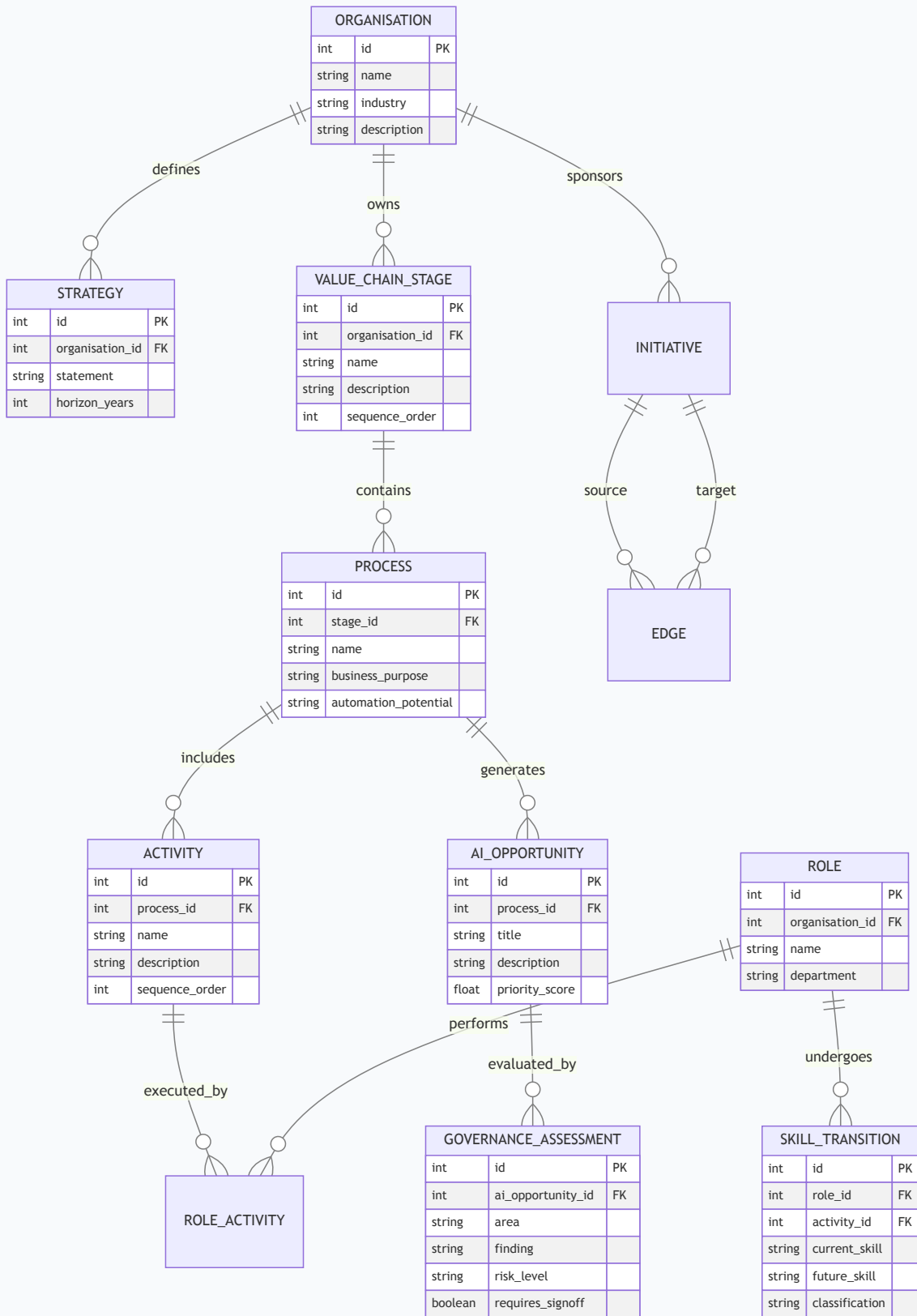
Enterprise AI Digital Twin - Database & Data Model Specification

Executive Summary

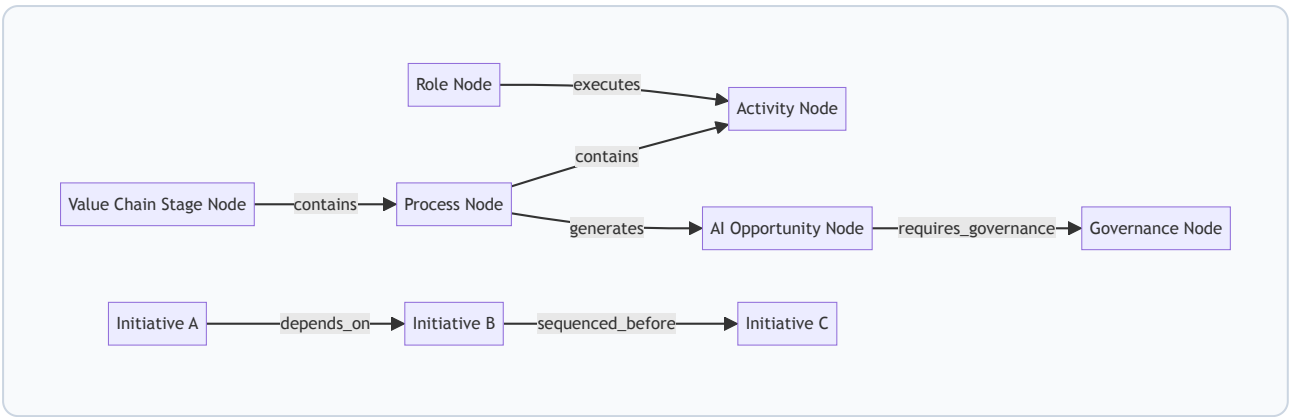
This document provides a comprehensive technical reference for the data persistence layer of the **Enterprise AI Digital Twin Platform**. The data architecture integrates three complementary storage paradigms:

- PostgreSQL Relational Database:** Structured entity storage for enterprise organizations, strategies, processes, activities, roles, AI opportunities, and compliance findings.
 - NetworkX Knowledge Graph Engine:** In-memory multi-hop directed graph (`G_live`) representing enterprise operational dependencies, sequencing, and dynamic simulation diffs.
 - Qdrant Vector Database:** 1536-dimensional vector embedding collections (`evidence_chunks` , `governance_sources`) powering semantic Retrieval-Augmented Generation (RAG).
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1. Entity-Relationship (ER) Diagram



2. NetworkX Knowledge Graph Model



Node Schema & Attributes

Node Type	Key Identifier Format	Primary Attributes
value_chain_stage	stage:{id}	name , description , sequence_order
process	process:{id}	name , business_purpose , automation_potential
activity	activity:{id}	name , sequence_order
role	role:{id}	name , department
ai_opportunity	opportunity:{id}	title , priority_score , business_benefit
initiative	initiative:{id}	name , priority_score

Directed Edge Classifications

- contains** : Parent-child structural containment (e.g., Stage → Process, Process → Activity).
- executes** : Role assignment to operational activities (Role → Activity).
- applies_to** : AI Opportunity relevance to processes/activities.
- requires_governance** : Links AI Opportunities to compliance assessments.
- depends_on** : Directed strategic dependency between initiatives.
- sequenced_before** : Temporal ordering constraint between initiatives.

3. Qdrant Vector Database Schema

Qdrant manages 1536-dimensional vector embeddings generated via `simple_embedding` / OpenAI embeddings for semantic evidence matching and governance citations.

3.1 `evidence_chunks` Collection

- **Vector Configuration:** 1536 Dimensions, Cosine Distance.
- **Purpose:** Stores benchmark transformation reports and empirical ROI evidence.
- **Payload Fields:**

```
{
  "source_url": "string (URL)",
  "title": "string",
  "entity_type": "string (e.g., 'ai_opportunity')",
  "entity_id": "integer",
  "content": "string (Text Content)"
}
```

3.2 `governance_sources` Collection

- **Vector Configuration:** 1536 Dimensions, Cosine Distance.
- **Purpose:** Stores regulatory frameworks (EU AI Act, DPDP Act 2023, NIST AI RMF 1.0, OECD AI Principles, ISO 42001).
- **Payload Fields:**

```
{
  "title": "string",
  "source_type": "enum ('law_regulation', 'regulatory_guidance', 'industry_standards')",
  "citation": "string (Article/Section reference)",
  "content": "string",
  "area": "enum ('privacy', 'human_oversight', 'model_risk', 'explainability', 'transparency')",
}
```

4. Relational Database Table Specifications

Core Tables Summary

1. **organisations** : Core enterprise entity (`id` , `name` , `industry` , `description`).
2. **strategies** : Strategic directives (`id` , `organisation_id` , `statement` , `horizon_years`).

3. **value_chain_stages** : Macro operational divisions (`id` , `organisation_id` , `name` , `description` , `sequence_order`).
4. **processes** : Core business processes (`id` , `stage_id` , `name` , `business_purpose` , `automation_potential`).
5. **activities** : Discrete operational steps (`id` , `process_id` , `name` , `description` , `sequence_order`).
6. **roles** : Job roles (`id` , `organisation_id` , `name` , `department`).
7. **role_activities** : Many-to-many junction table mapping Roles to Activities.
8. **skill_transitions** : Skill evolution records (`id` , `role_id` , `activity_id` , `current_skill` , `future_skill` , `classification`).
9. **ai_opportunities** : Candidate AI automation projects (`id` , `process_id` , `title` , `description` , `priority_score`).
10. **governance_assessments** : Compliance evaluation records (`id` , `ai_opportunity_id` , `area` , `finding` , `risk_level` , `requires_signoff`).
11. **initiatives** : Strategic initiatives (`id` , `organisation_id` , `name` , `description` , `priority_score`).
12. **edges** : Directed dependency graph links between initiatives (`id` , `source_type` , `source_id` , `target_type` , `target_id` , `relationship_type`).